Web API Development with ASP.NET Core (.NET 10)

Final Assessment - Multiple Choice Questions

Name:		**Date:**	
Batch / Roll No.:		**Score:**	_____ / 20

Total questions	20 (Day 1: 5, Day 2: 7, Day 3: 8)
Marks	1 mark per question, total 20. No negative marking.
Suggested time	30 minutes
Level	Fresher

Instructions: Each question has exactly one correct answer. Read every option carefully and choose the best answer. Write the letter (A, B, C or D) of your answer next to the question number on the answer sheet or circle the option as instructed by the examiner.

SECTION A - Day 1: API Foundations, REST and Dependency Injection

Q1. Which HTTP method is normally used to create a new resource, and which status code should the API return when the creation succeeds?

A) GET, with 200 OK
B) PUT, with 204 No Content
C) POST, with 201 Created
D) DELETE, with 202 Accepted

Q2. Which endpoint follows REST naming guidelines for getting the list of all students?

A) GET /api/students
B) GET /api/getAllStudents
C) POST /api/getstudents
D) GET /api/students/getlist

Q3. An HTTP method is called idempotent. What does this mean?

A) The method never reads any data
B) The response is always cached by the browser
C) The method can be used only once per user session
D) Sending the same request once or many times leaves the server in the same final state

Q4. In ASP.NET Core, a service registered with the Scoped lifetime is created:

A) Every time it is requested, even inside the same request
B) Once per HTTP request and shared inside that request
C) Once for the whole life of the application
D) Only when the application is compiled

Q5. Why do we return DTOs from an API instead of returning database entity classes directly?

A) To control exactly which data is exposed and keep the API shape independent from the database model
B) Because entity classes cannot be converted to JSON
C) Because Swagger refuses to run with entity classes
D) Because DTOs automatically create database tables

SECTION B - Day 2: EF Core, Validation and Error Handling

Q6. After adding a new property to an entity class, which commands apply this change to the SQL Server database using EF Core migrations?

A) dotnet build, then dotnet publish
B) dotnet ef database drop only
C) dotnet ef dbcontext scaffold, then dotnet run
D) dotnet ef migrations add NAME, then dotnet ef database update

Q7. When should you use `AsNoTracking()` in an EF Core query?

A) Whenever you plan to change the entity and call SaveChangesAsync
B) For read-only queries where the returned entities will not be updated
C) To make EF Core create the table automatically
D) To load related entities with a JOIN

Q8. In EF Core, when are the changes made with `Add`, `Update` and `Remove` actually written to the database?

A) Immediately when Add is called
B) When the controller action starts
C) When SaveChangesAsync is called
D) When DbContext is registered in Program.cs

Q9. A Course has many Students and each Student refers back to its Course. Returning the Course entity together with its Students causes a "possible object cycle" JSON error. What is the best fix?

A) Return a DTO created with Select instead of returning the entity objects
B) Register DbContext as a Singleton
C) Remove the foreign key column from the database
D) Call SaveChangesAsync before returning the data

Q10. A controller has `[ApiController]` and the request DTO has `[Required]` on Email. The client sends JSON without Email. What happens by default?

A) The action runs and returns 200 OK
B) ASP.NET Core returns 404 Not Found
C) ASP.NET Core returns 500 Internal Server Error
D) ASP.NET Core returns 400 Bad Request before the action method runs

Q11. Which statement about a global exception handler (`IExceptionHandler`) in production is correct?

A) It should return the full stack trace so users can fix the problem
B) It should log the full exception and return a safe ProblemDetails response with a trace id, without the stack trace
C) It should convert every exception into 200 OK
D) It is required to make [Required] validation work

Q12. Why should a paged query use a stable ordering (for example OrderBy followed by ThenBy on Id) before Skip and Take?

A) To automatically increase the page size
B) To make EF Core track the entities
C) Without a defined order, the same row may appear on two pages or a row may be missing
D) To convert the query into a stored procedure

SECTION C - Day 3: Security, Testing and Production Readiness

Q13. A request contains a valid token, but the signed-in user is not allowed to perform the action. Which status code should the API return?

A) 403 Forbidden
B) 401 Unauthorized
C) 404 Not Found
D) 400 Bad Request

Q14. How should a user's password be stored in the database?

A) As plain text so that administrators can recover it
B) As a salted hash created with a password hasher such as PasswordHasher
C) As Base64 encoded text
D) Inside the JWT payload

Q15. Which statement about a JWT is correct?

A) The payload is encrypted, so secrets can safely be stored in it
B) The signature hides the payload from the client
C) A JWT can never expire
D) The signature helps detect changes to the token, but the payload can still be read by anyone who has the token

Q16. Which is the correct order of these middleware calls in Program.cs?

A) UseAuthorization() first, then UseAuthentication()
B) Only UseAuthorization() is needed because it also validates the token
C) UseAuthentication() first, then UseAuthorization()
D) Both can be removed if [Authorize] is placed on a controller

Q17. What is the purpose of a CORS policy on an API?

A) To tell browsers which other origins are allowed to call the API and read its responses
B) To encrypt the traffic between client and API
C) To check the user's password
D) To limit the number of requests per minute

Q18. A fixed-window rate limiter allows 3 requests per 30 seconds. What should the 4th request in the same window receive?

A) 200 OK
B) 429 Too Many Requests
C) 401 Unauthorized
D) 500 Internal Server Error

Q19. Which statement best describes an integration test written with `WebApplicationFactory<Program>`?

A) It tests one method in isolation without HTTP or middleware
B) It builds a Docker image of the API
C) It replaces Swagger and Postman in production
D) It starts the API in memory and sends real HTTP requests through the pipeline, for example to check that a protected endpoint returns 401

Q20. Which kind of failure is the best candidate for an automatic retry with Polly?

A) A wrong password that returns 401
B) A validation error that returns 400
C) A temporary failure such as a short network problem or a 503 Service Unavailable response
D) A missing resource that returns 404

ANSWER KEY

For examiner use only - remove this page before giving the question paper to students.

Q	Answer	Topic	Why this is correct
1	**C**	Day 1 - 1.1 HTTP and REST	POST creates a resource and a successful creation returns 201 Created (usually with a Location header).
2	**A**	Day 1 - 1.4 API design	Use plural nouns for resources and let the HTTP method express the action. Verbs in the URL are avoided.
3	**D**	Day 1 - 1.1 HTTP and REST	Idempotent means repeating the same request does not change the final state further (for example PUT and DELETE).
4	**B**	Day 1 - 1.8 Dependency Injection	Scoped = one instance per request. Transient = new every time. Singleton = one for the whole application.
5	**A**	Day 1 - 1.6 Model binding and DTOs	DTOs expose only the required fields and protect the API contract from database changes.
6	**D**	Day 2 - 2.2 Migrations	migrations add creates the migration file; database update applies it to the database.
7	**B**	Day 2 - 2.1 EF Core fundamentals	AsNoTracking skips change tracking, which is faster and suitable for read-only data.
8	**C**	Day 2 - 2.3 Async CRUD	Add, Update and Remove only mark changes. SaveChangesAsync sends them to the database.
9	**A**	Day 2 - 2.4 Relationships	DTO projection returns only the needed fields and removes the circular reference.
10	**D**	Day 2 - 2.5 Validation	[ApiController] enables automatic model validation and returns 400 when the model is invalid.
11	**B**	Day 2 - 2.7 Exception handling	Users get a safe message and a trace id; developers read the details in the logs.
12	**C**	Day 2 - 2.8 Paging	Stable ordering gives predictable pages. Without it the database may return rows in a different order each time.
13	**A**	Day 3 - 3.1 Security essentials	401 means the identity is missing or invalid. 403 means the identity is valid but the permission is missing.
14	**B**	Day 3 - 3.1 Security essentials	Passwords are never stored in plain text. A salted one-way hash is stored and compared during login.
15	**D**	Day 3 - 3.1 / 3.2 JWT	A JWT is signed, not encrypted. Do not put passwords or secrets in the payload. Tokens carry an expiry time.
16	**C**	Day 3 - 3.2 JWT authentication	The API must first know who you are (authentication) and then decide what you may do (authorization).
17	**A**	Day 3 - 3.4 CORS and configuration	CORS is a browser rule. It does not encrypt traffic or authenticate users.
18	**B**	Day 3 - 3.6 Rate limiting	429 is the standard status code when a client sends too many requests.
19	**D**	Day 3 - 3.9 Integration tests	Integration tests check that routing, middleware, authentication and controllers work together.
20	**C**	Day 3 - 3.10 Polly	Retry helps only with temporary (transient) faults. Permanent errors will fail again.

Quick key: 1-C | 2-A | 3-D | 4-B | 5-A | 6-D | 7-B | 8-C | 9-A | 10-D | 11-B | 12-C | 13-A | 14-B | 15-D | 16-C | 17-A | 18-B | 19-D | 20-C